Calculus I

Tutorial #2 (7-11 Oct 2024)

Topics:

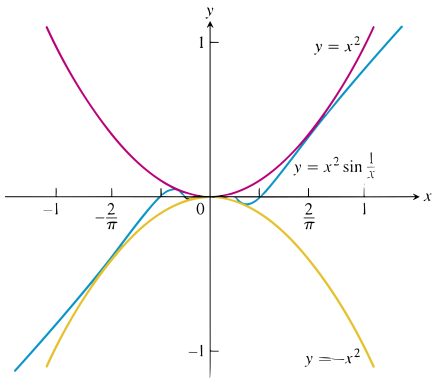
- Definition of limits
- Infinite limits
- Continuity

Problem 1. Prove by definition of limit that $\lim_{x\to 1} f(x) = 1$ if

$$f(x) = \begin{cases} x^2, & x \neq 1, \\ 2, & x = 1. \end{cases}$$

Problem 2. *(Squeeze theorem)* Prove that

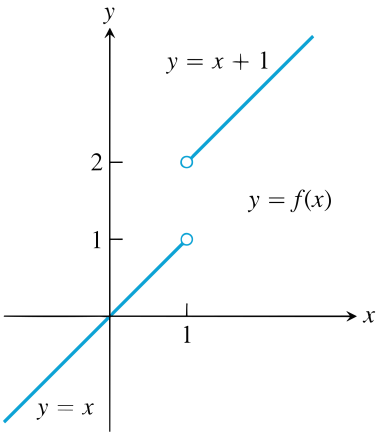
$$\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0.$$



Problem 3. *(One-sided limits)* Let

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x > 1 \end{cases}$$

Prove that $\lim_{x\to 1} f(x)$ does not exist.

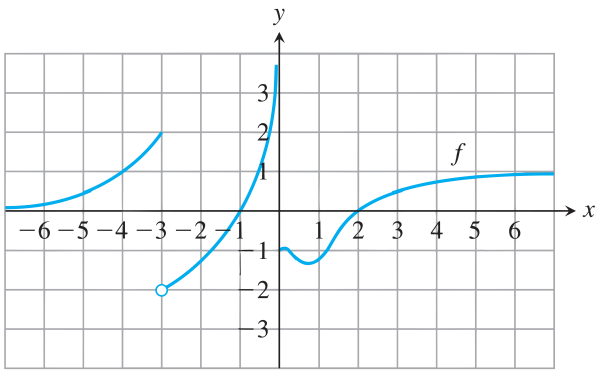


Problem 4. For the function f whose graph is given, determine the following limits.

a) $\lim_{x\to 2} f(x)$; **b)** $\lim_{x\to -3^+} f(x)$; **c)** $\lim_{x\to -3^-} f(x)$.

d) $\lim_{x\to -3} f(x)$; **e)** $\lim_{x\to 0^+} f(x)$; **f)** $\lim_{x\to 0^-} f(x)$.

g) $\lim_{x\to 0} f(x)$; **h)** $\lim_{x\to \infty} f(x)$; **i)** $\lim_{x\to -\infty} f(x)$.



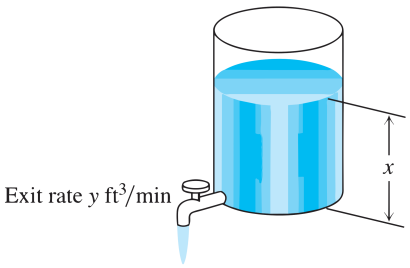
Problem 5. *(Thermal expansion in precise equipment)* As you may know, most metals expand when heated and contract when cooled. The dimensions of a piece of laboratory equipment are sometimes so critical that the shop where the equipment is made must be held at the same temperature as the laboratory where the equipment is to be used. A typical aluminum bar that is 10 cm wide at $70°F$ will be

$$y = 10 + (t - 70) \times 10^{-4}$$

centimeters wide at a nearby temperature t. Suppose that you are using a bar like this in a gravity wave detector, where its width must stay within 0.0005 cm of the ideal 10 cm. How close to $t_0 = 70°F$ must you maintain the temperature to ensure that this tolerance is not exceeded?

Problem 6. *(Controlling the flow from a draining tank)* Torricelli's law says that if you drain a tank like the one in the figure shown, the rate y at which water runs out is a constant times the square root of the water's depth x. The constant depends on

the size and shape of the exit valve.



Suppose that $y = \sqrt{x}/2$ for a certain tank. You are trying to maintain a fairly constant exit rate by adding water to the tank with a hose from time to time. How deep must you keep the water if you want to maintain the exit rate

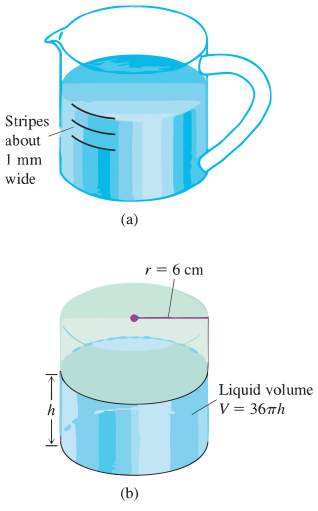
a. within $0.2\, ft^3/min$ of the rate $y_0 = 1\, ft^3/min$?

b. within $0.1\, ft^3/min$ of the rate $y_0 = 1\, ft^3/min$?

Problem 7. *(Stripes on a measuring cup)* The interior of a typical $1\,L$ measuring cup is a right circular cylinder of radius $6cm$ (see accompanying figure). The volume of water we put in the cup is therefore a function of the level h to which the cup is filled, the formula being

$$V = \pi 6^2 h = 36\pi h.$$

How closely must we measure h to measure out $1\,L$ of water ($1000\,cm^3$) with an error of no more than 1% ($10\,cm^3$)?



A 1-L measuring cup (a), modeled as a right circular cylinder (b) of radius $r = 6$ cm

Problem 8. For what value of a is

$$f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$$

continuous at every x?

Problem 9. If the product function $h(x) = f(x) \cdot g(x)$ is continuous at $x = 0$ must $f(x)$ and $g(x)$ be continuous at $x = 0$? Justify your answer.

Problem 10. *(Bounded functions)* A real-valued function f is **bounded from above** on a set D if there exists a number N such that $f(x) \leq N$ for all x in D. We call N, when it exists, an **upper bound** for f on D and say that f is bounded from above by N. In a similar manner, we say that f is **bounded from below** on D if there exists a number M such that $f(x) \geq M$ for all x in D. We call M, when it exists, a **lower bound** for f on D and say that f is bounded from below by M. We say that f is **bounded** on D if it is bounded from both above and below.

(a) Show that f is bounded on D if and only if there exists a number B such that $|f(x)| \leq B$ for all x in D.

(b) Suppose that f is bounded from above by N. Show that if $\lim_{x \to x_0} f(x) = L$, then $L \leq N$.

(c) Suppose that f is bounded from below by M. Show that if $\lim_{x \to x_0} f(x) = L$, then $L \geq M$.

THE END

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Answer to selected problems

4. a) 0; b) -2; c) 2; d) limit does not exist; e) -1; f) $+\infty$; g) the limit does not exist; h) 1; i) 0.

8. $\frac{4}{3}$

9. False.